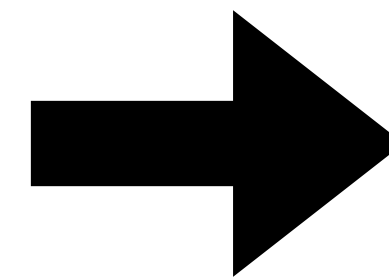
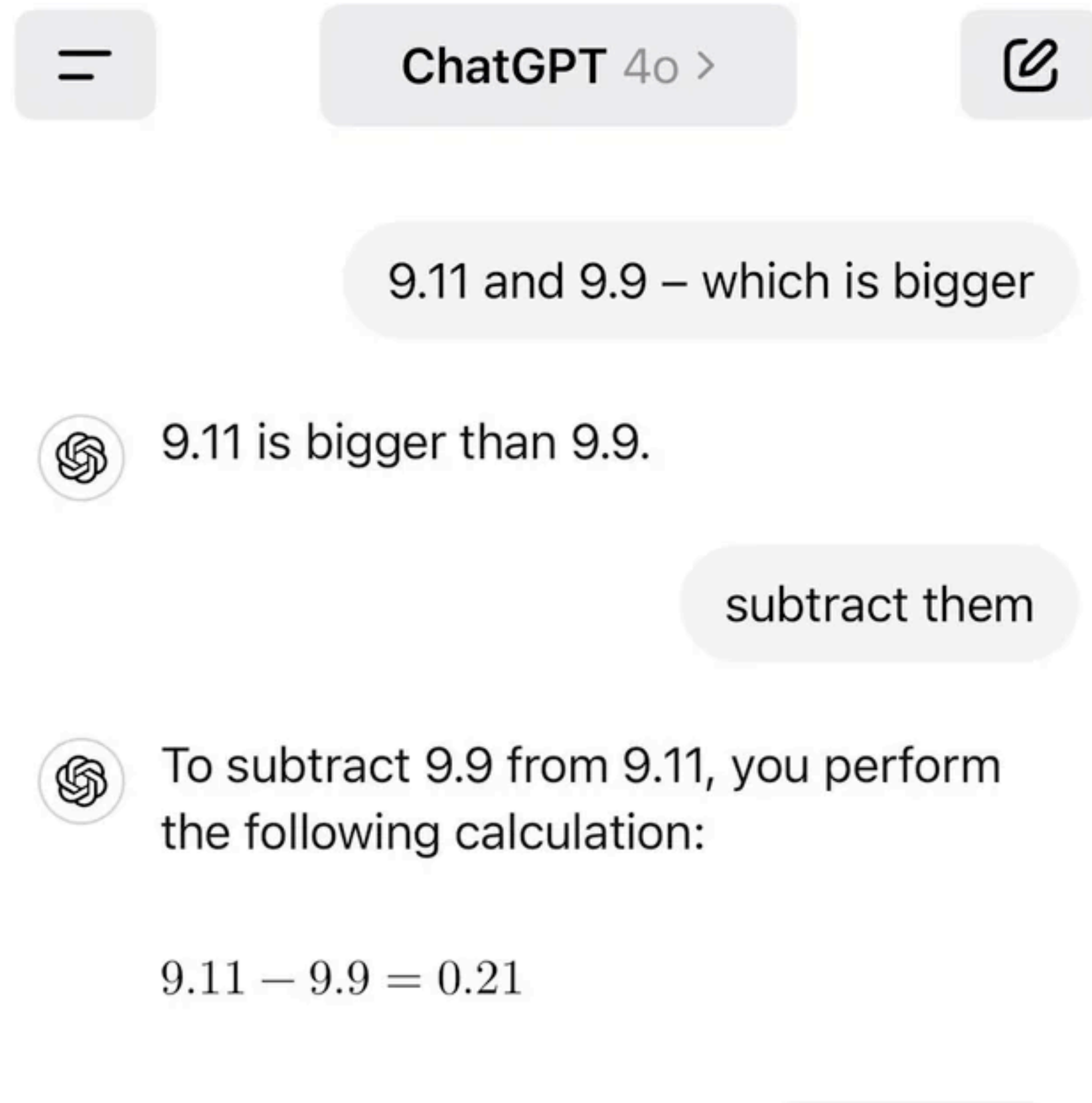


How AI can solve my math hw?

about LEAN

윤승현

Q: How it can win Olympiad?



magic?

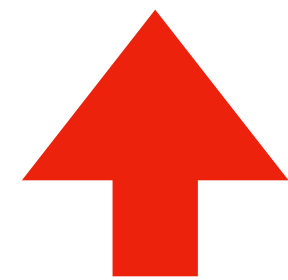


Hallucination

hw #1: Pigeons in holes. Here there are 10 pigeons and 9 holes.

Is there any holes has more than one pigeon?

(ChatGPT): YES! it's obvious!



Give me proof, sir





it's time to LEAN!

hw #1: Pigeons in holes. Here there are 10 pigeons and 9 holes.

Is there any holes has more than one pigeon?

assume there's no such hole

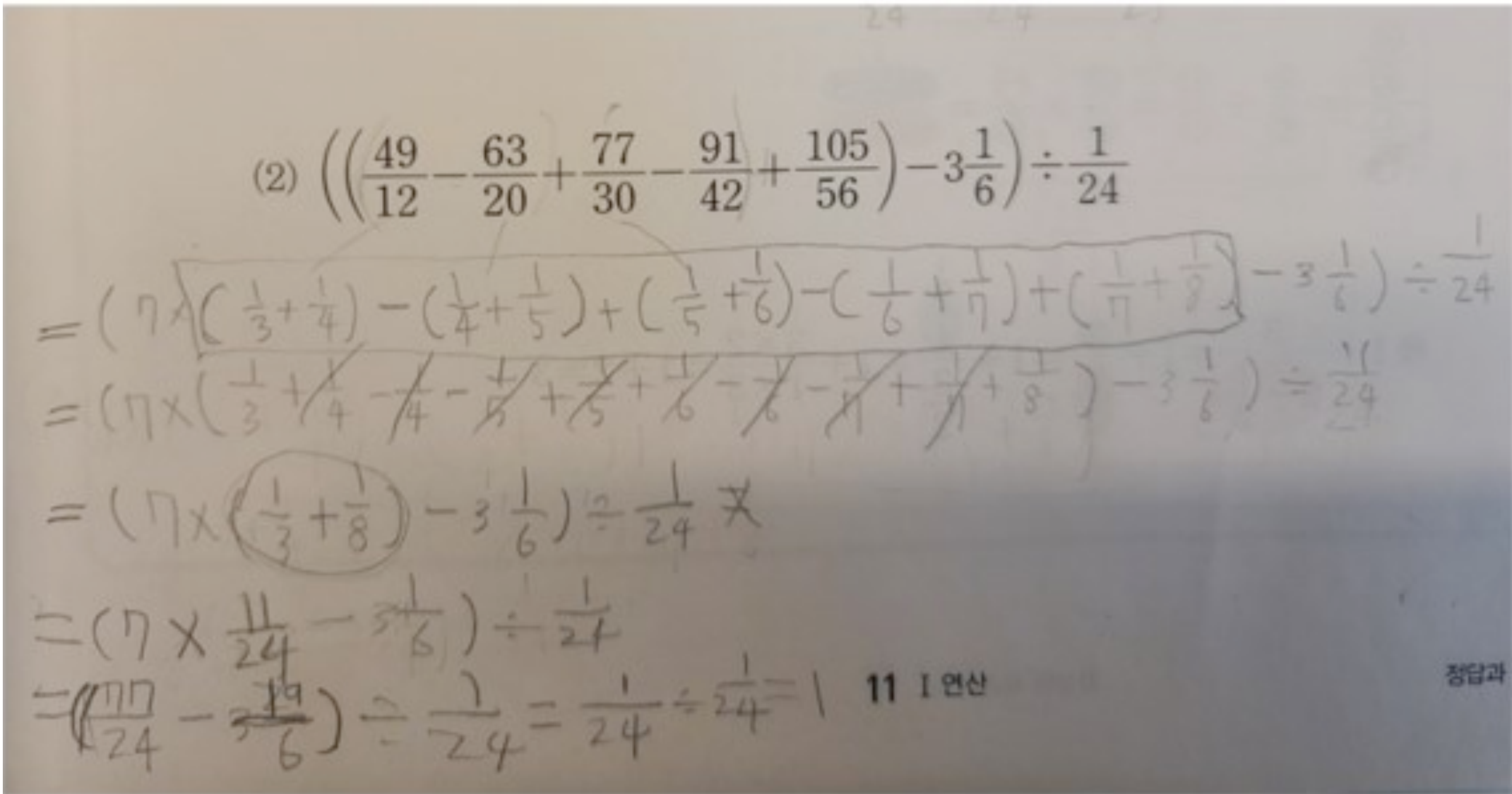
all of holes have at most 1 pigeons

the number of pigeons is less than $9 * 1$

it is contradiction

```
17 theorem pigeon_hole_principle (counts: List ℕ) (h_holes: counts.length = 9)
   (h_pigeons: counts.sum = 10) ∃ m ∈ counts, m ≥ 2 := by
18   by_contra! h_all_lt_2
19   have h_all_le_1 : ∀ m ∈ counts, m ≤ 1 := by
20     intro m hm_in
21     exact Nat.le_of_lt_succ (h_all_lt_2 m hm_in)
22
23   have h_sum_le_9 : counts.sum ≤ counts.length :=
24     sum_le_length counts h_all_le_1
25
26   rw [h_pigeons, h_holes] at h_sum_le_9
27   norm_num at h_sum_le_9
```


it's time to LEAN!



verify time 💰



no ambiguity(=trustworthy)

Thank you!